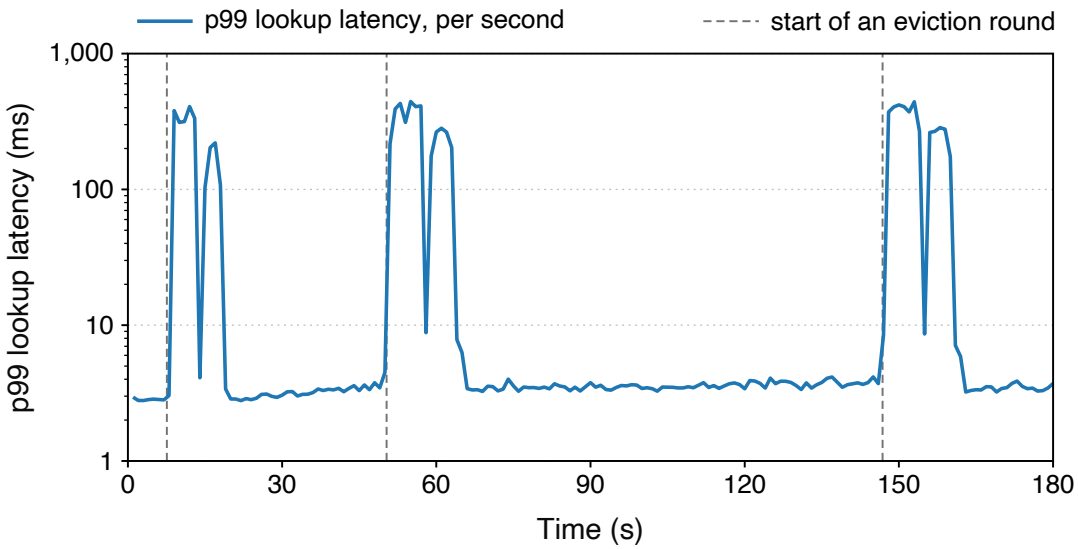
Experiment 2: Full pool. Eviction stalls lookups for 10–15 s



- **Setup.** 100M objects of 1 MiB at the 90% eviction watermark, so every write forces eviction. 16 clients write; 16 look up recent objects.
- **Eviction** (Mooncake defaults). Each round evicts 5% of objects after scanning all keys twice, holding shard locks.
- **Result.** Between rounds p99 is 3.5 ms. At each round start it reaches 220–440 ms for 10–15 s, and writes stop.
- **Fixable, but shared.** Incremental eviction would remove the scan; in a central master, any such background work still stalls every node at once.

One 180 s run; p99 per second; dashed lines: round starts from the master's log. 6% of writes were rejected for lack of space.